

AdaptiveAvgPool2d#

<https://pytorch.org/docs/stable/generated/torch.nn.AdaptiveAvgPool2d.html>

Description:

Applies a 2D adaptive average pooling over an input signal composed of several input planes. The output is of size H x W, for any input size. The number of output features is equal to the number of input planes.

`output_size` (Union[int, None, tuple[Optional[int], Optional[int]]]) – the target output size of the image of the form H x W. Can be a tuple (H, W) or a single H for a square image H x H. H and W can be either a int, or None which means the size will be the same as that of the input. Input: (N,C,Hin,Win)(N, C, H_{in}, W_{in})(N,C,Hin,Win) or (C,Hin,Win)(C, H_{in}, W_{in})(C,Hin,Win). Output: (N,C,S0,S1)(N, C, S_{0}, S_{1})(N,C,S0,S1) or (C,S0,S1)(C, S_{0}, S_{1})(C,S0,S1), where $S=\text{output_size}$.

Function Signature

```
class torch.nn.AdaptiveAvgPool2d(output_size)[source]#
```

Parameters

Shape:

```
forward(input)[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> # target output size of 5x7
>>> m = nn.AdaptiveAvgPool2d((5, 7))
>>> input = torch.randn(1, 64, 8, 9)
>>> output = m(input)
>>> # target output size of 7x7 (square)
>>> m = nn.AdaptiveAvgPool2d(7)
>>> input = torch.randn(1, 64, 10, 9)
>>> output = m(input)
>>> # target output size of 10x7
>>> m = nn.AdaptiveAvgPool2d((None, 7))
>>> input = torch.randn(1, 64, 10, 9)
>>> output = m(input)
```

Detailed Documentation

Documentation

Applies a 2D adaptive average pooling over an input signal composed of several input planes.

The output is of size $H \times W$, for any input size. The number of output features is equal to the number of input planes.

`output_size` (`Union[int, None, tuple[Optional[int], Optional[int]]]`) – the target output size of the image of the form $H \times W$. Can be a tuple (H, W) or a single H for a square image $H \times H$. H and W can be either a `int`, or `None` which means the size will be the same as that of the input.

Input: (N, C, H_{in}, W_{in}) or (N, C, H_{in}, W_{in}) or (C, H_{in}, W_{in}) or (C, H_{in}, W_{in}) .

Output: (N, C, S_0, S_1) or (N, C, S_0, S_1) or (C, S_0, S_1) or (C, S_0, S_1) , where $S = \text{output_size}$.

Runs the forward pass.